

Modern Practical Control Systems

Chapter 0

Mathematical Foundations

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Contents

1	Mathematical Foundations	5
1.1	Complex Numbers and Imaginary Arithmetic	5
1.1.1	Definition and Representation	6
1.1.2	Arithmetic Operations	7
1.1.3	Complex Numbers in Control Systems	7
1.2	Calculus Review	8
1.2.1	Differential Calculus	8
1.2.2	Integral Calculus	9
1.3	Differential Equations	10
1.3.1	Ordinary Differential Equations (ODEs)	10
1.3.2	Systems of First-Order ODEs	12
1.4	Laplace Transforms	12
1.4.1	Definition	12
1.4.2	Common Laplace Transform Pairs	14
1.4.3	Laplace Transform Properties	14
1.4.4	Partial Fraction Expansion	14
1.4.5	Transfer Functions	14
1.5	Fourier Transform and Frequency Analysis	15
1.5.1	Fourier Series	15
1.5.2	Fourier Transform	15
1.5.3	Frequency Response	16
1.5.4	Bode Plots	16
1.6	Linear Algebra	16
1.6.1	Vectors and Matrices	16
1.6.2	Special Matrices	18
1.6.3	Determinant and Rank	18
1.6.4	Eigenvalues and Eigenvectors	18
1.6.5	Matrix Decompositions	19
1.6.6	Matrix Exponential	19
1.6.7	Linear Systems of Equations	20
1.7	Probability and Statistics	20
1.7.1	Probability Fundamentals	20
1.7.2	Random Variables	21
1.7.3	Common Distributions	21
1.7.4	Stochastic Processes	21
1.7.5	Estimation Theory	22
1.8	Optimization	22

1.8.1	Unconstrained Optimization	22
1.8.2	Constrained Optimization	22
1.8.3	Convex Optimization	22
1.8.4	Dynamic Programming	23
1.9	Discrete-Time Mathematics	23
1.9.1	Sequences and Difference Equations	23
1.9.2	The Z-Transform	23
1.9.3	Relationship Between s-Domain and z-Domain	24
1.9.4	Discrete-Time State-Space	25
1.10	Special Functions and Signals	25
1.10.1	Unit Step Function	25
1.10.2	Impulse (Dirac Delta) Function	25
1.10.3	Ramp Function	25
1.10.4	Sinusoidal Functions	26
1.10.5	Exponential Functions	26
1.11	Numerical Methods	26
1.11.1	Numerical Integration of ODEs	26
1.11.2	Root Finding	27
1.11.3	Numerical Linear Algebra	27
1.12	Chapter Summary	27
1.13	Exercises	27

Chapter 1

Mathematical Foundations

Learning Objectives

After completing this chapter, you will be able to:

- Perform arithmetic and algebraic operations with complex numbers
- Apply calculus techniques including differentiation, integration, and differential equations
- Use Laplace and Fourier transforms to analyze systems
- Perform matrix operations and understand eigenvalue analysis
- Apply probability and statistics concepts relevant to stochastic control and machine learning
- Understand optimization fundamentals for control design
- Work with discrete-time mathematics for digital control systems

Control systems engineering draws upon a rich mathematical foundation spanning calculus, linear algebra, complex analysis, differential equations, and increasingly, probability theory and optimization. This chapter provides a comprehensive review of these mathematical tools, ensuring all readers have the prerequisite knowledge for the chapters that follow.

Whether you are approaching control from a mechanical, electrical, chemical, biomedical, aerospace, or computer science background, mastery of these fundamentals will enable you to understand system behavior, design controllers, and implement modern machine learning-based control strategies.

1.1 Complex Numbers and Imaginary Arithmetic

Complex numbers are fundamental to control systems analysis. They appear naturally in the solution of differential equations, in the Laplace transform, in frequency response analysis, and in stability analysis.

1.1.1 Definition and Representation

Definition 0.1: Complex Number

A **complex number** z is defined as:

$$z = a + jb \quad (1.1)$$

where $a, b \in \mathbb{R}$, and $j = \sqrt{-1}$ is the **imaginary unit**. The real part is $\text{Re}(z) = a$ and the imaginary part is $\text{Im}(z) = b$.

Note 1.1. In electrical engineering, j is often used instead of i for the imaginary unit to avoid confusion with current. We adopt this convention throughout.

Complex numbers can be represented in multiple forms:

Rectangular Form:

$$z = a + jb \quad (1.2)$$

Polar Form:

$$z = r(\cos \theta + j \sin \theta) = r \angle \theta \quad (1.3)$$

where $r = |z| = \sqrt{a^2 + b^2}$ is the **magnitude** (or modulus) and $\theta = \arg(z) = \arctan(b/a)$ is the **argument** (or phase angle).

Exponential Form (Euler's Formula):

$$z = re^{j\theta} \quad (1.4)$$

Euler's formula, one of the most elegant results in mathematics, states:

$$e^{j\theta} = \cos \theta + j \sin \theta \quad (1.5)$$

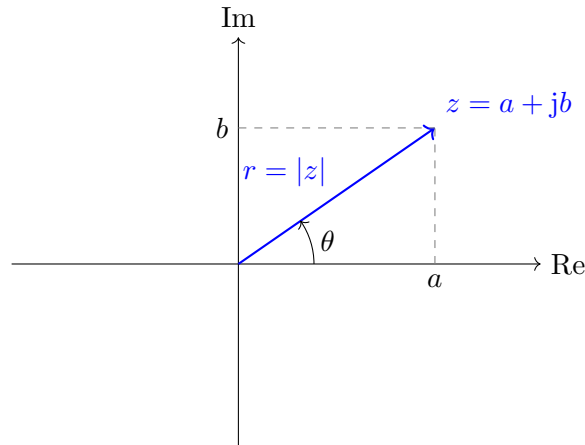


Figure 1.1: Geometric representation of a complex number in the complex plane (Argand diagram).

1.1.2 Arithmetic Operations

Addition and Subtraction:

$$(a + jb) \pm (c + jd) = (a \pm c) + j(b \pm d) \quad (1.6)$$

Multiplication:

$$(a + jb)(c + jd) = (ac - bd) + j(ad + bc) \quad (1.7)$$

In polar form: $r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$

Division:

$$\frac{a + jb}{c + jd} = \frac{(a + jb)(c - jd)}{(c + jd)(c - jd)} = \frac{(ac + bd) + j(bc - ad)}{c^2 + d^2} \quad (1.8)$$

In polar form: $\frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$

Complex Conjugate:

$$z^* = \bar{z} = a - jb \quad (1.9)$$

Important property: $z \cdot z^* = |z|^2 = a^2 + b^2$

Powers (De Moivre's Theorem):

$$z^n = r^n e^{jn\theta} = r^n (\cos n\theta + j \sin n\theta) \quad (1.10)$$

Roots: The n th roots of $z = re^{j\theta}$ are:

$$z^{1/n} = r^{1/n} e^{j(\theta + 2\pi k)/n}, \quad k = 0, 1, \dots, n-1 \quad (1.11)$$

Example 0.1: Complex Number Operations

Find the polar form, magnitude, and phase of $z = 3 + 4j$.

Solution:

$$|z| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5 \quad (1.12)$$

$$\theta = \arctan\left(\frac{4}{3}\right) \approx 53.13 = 0.927 \text{ rad} \quad (1.13)$$

$$z = 5e^{j(0.927)} = 5\angle 53.13 \quad (1.14)$$

1.1.3 Complex Numbers in Control Systems

Complex numbers appear throughout control theory:

- **Poles and Zeros:** The roots of transfer function polynomials are generally complex.
- **Eigenvalues:** System matrices often have complex eigenvalues that determine stability and oscillatory behavior.
- **Frequency Response:** Evaluating transfer functions at $s = j\omega$ yields complex numbers representing magnitude and phase.
- **Sinusoidal Steady-State:** Complex exponentials simplify analysis of sinusoidal inputs.

1.2 Calculus Review

1.2.1 Differential Calculus

Derivatives and Rules

The derivative of a function $f(t)$ represents its instantaneous rate of change:

$$\frac{df}{dt} = f'(t) = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} \quad (1.15)$$

Key Differentiation Rules:

Rule	Formula
Power Rule	$\frac{d}{dt}(t^n) = nt^{n-1}$
Exponential	$\frac{d}{dt}(e^{at}) = ae^{at}$
Logarithm	$\frac{d}{dt}(\ln t) = \frac{1}{t}$
Sine	$\frac{d}{dt}(\sin \omega t) = \omega \cos \omega t$
Cosine	$\frac{d}{dt}(\cos \omega t) = -\omega \sin \omega t$
Product Rule	$\frac{d}{dt}(fg) = f'g + fg'$
Quotient Rule	$\frac{d}{dt}\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}$
Chain Rule	$\frac{d}{dt}f(g(t)) = f'(g(t)) \cdot g'(t)$

Table 1.1: Essential differentiation rules for control systems.

Partial Derivatives

For functions of multiple variables $f(x_1, x_2, \dots, x_n)$, partial derivatives hold all variables constant except one:

$$\frac{\partial f}{\partial x_i} = \lim_{\Delta x_i \rightarrow 0} \frac{f(x_1, \dots, x_i + \Delta x_i, \dots, x_n) - f(x_1, \dots, x_n)}{\Delta x_i} \quad (1.16)$$

The **gradient** is the vector of all partial derivatives:

$$\nabla f = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \quad \dots \quad \frac{\partial f}{\partial x_n} \right]^\top \quad (1.17)$$

The **Hessian** is the matrix of second partial derivatives:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad (1.18)$$

Taylor Series Expansion

Taylor series are essential for linearization of nonlinear systems:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots \quad (1.19)$$

For multivariate functions about point $\mathbf{x}_0$:

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^\top \mathbf{H}(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0) \quad (1.20)$$

Linearization in Control

For nonlinear systems $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$, we linearize about an equilibrium point $(\mathbf{x}_0, \mathbf{u}_0)$:

$$\Delta \dot{\mathbf{x}} \approx \mathbf{A} \Delta \mathbf{x} + \mathbf{B} \Delta \mathbf{u} \quad (1.21)$$

where $\mathbf{A} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{(\mathbf{x}_0, \mathbf{u}_0)}$ and $\mathbf{B} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_{(\mathbf{x}_0, \mathbf{u}_0)}$

1.2.2 Integral Calculus

Definite and Indefinite Integrals

The indefinite integral (antiderivative):

$$\int f(t) dt = F(t) + C \quad \text{where} \quad F'(t) = f(t) \quad (1.22)$$

The definite integral:

$$\int_a^b f(t) dt = F(b) - F(a) \quad (1.23)$$

Key Integration Formulas:

Function	Integral
$t^n \ (n \neq -1)$	$\frac{t^{n+1}}{n+1} + C$
$\frac{1}{t}$	$\ln t + C$
e^{at}	$\frac{1}{a} e^{at} + C$
$\sin \omega t$	$-\frac{1}{\omega} \cos \omega t + C$
$\cos \omega t$	$\frac{1}{\omega} \sin \omega t + C$
$\frac{1}{a^2 + t^2}$	$\frac{1}{a} \arctan \frac{t}{a} + C$

Table 1.2: Essential integration formulas.

Integration by Parts

$$\int u dv = uv - \int v du \quad (1.24)$$

This technique is essential for deriving Laplace transform properties.

Convolution Integral

The convolution of two functions $f(t)$ and $g(t)$ is:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau \quad (1.25)$$

For causal systems (signals zero for $t < 0$):

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau \quad (1.26)$$

Convolution in Control Systems

The output $y(t)$ of a linear time-invariant (LTI) system with impulse response $h(t)$ to input $u(t)$ is:

$$y(t) = (h * u)(t) = \int_0^t h(\tau)u(t - \tau) d\tau \quad (1.27)$$

1.3 Differential Equations

Differential equations describe how systems evolve over time and are the mathematical language of dynamics.

1.3.1 Ordinary Differential Equations (ODEs)

First-Order Linear ODEs

The general first-order linear ODE:

$$\frac{dy}{dt} + p(t)y = q(t) \quad (1.28)$$

Solution using Integrating Factor: Multiply by $\mu(t) = e^{\int p(t) dt}$:

$$y(t) = \frac{1}{\mu(t)} \left[\int \mu(t)q(t) dt + C \right] \quad (1.29)$$

For constant coefficients $\frac{dy}{dt} + ay = b$:

$$y(t) = \frac{b}{a} + \left(y_0 - \frac{b}{a} \right) e^{-at} \quad (1.30)$$

Example 0.2: RC Circuit

An RC circuit with voltage source V_s satisfies:

$$RC \frac{dv_c}{dt} + v_c = V_s \quad (1.31)$$

With $v_c(0) = 0$, the capacitor voltage is:

$$v_c(t) = V_s \left(1 - e^{-t/RC} \right) \quad (1.32)$$

The time constant $\tau = RC$ determines how quickly the system responds.

Second-Order Linear ODEs with Constant Coefficients

The standard form:

$$\frac{d^2y}{dt^2} + 2\zeta\omega_n\frac{dy}{dt} + \omega_n^2y = \omega_n^2u(t) \quad (1.33)$$

where:

- ω_n = natural frequency (rad/s)
- ζ = damping ratio (dimensionless)

The **characteristic equation** is:

$$\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2 = 0 \quad (1.34)$$

with roots:

$$\lambda_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \quad (1.35)$$

Cases based on damping ratio:

1. **Overdamped** ($\zeta > 1$): Two distinct real roots. Response: $y(t) = C_1e^{\lambda_1t} + C_2e^{\lambda_2t}$
2. **Critically Damped** ($\zeta = 1$): Repeated real root $\lambda = -\omega_n$. Response: $y(t) = (C_1 + C_2t)e^{-\omega_nt}$
3. **Underdamped** ($0 < \zeta < 1$): Complex conjugate roots $\lambda = -\zeta\omega_n \pm j\omega_d$ where $\omega_d = \omega_n\sqrt{1 - \zeta^2}$. Response:

$$y(t) = e^{-\zeta\omega_nt}(C_1 \cos \omega_dt + C_2 \sin \omega_dt) \quad (1.36)$$
4. **Undamped** ($\zeta = 0$): Pure imaginary roots. Response: $y(t) = C_1 \cos \omega_nt + C_2 \sin \omega_nt$

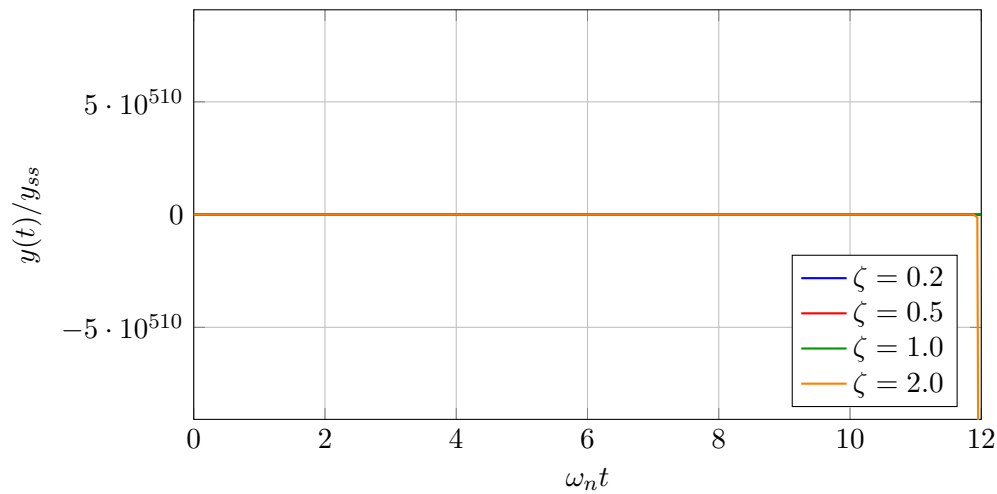


Figure 1.2: Step response of second-order systems for different damping ratios.

Higher-Order Linear ODEs

The general n th-order linear ODE with constant coefficients:

$$a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = f(t) \quad (1.37)$$

The solution consists of:

- **Homogeneous solution** $y_h(t)$: From the characteristic equation roots
- **Particular solution** $y_p(t)$: Depends on the forcing function $f(t)$

1.3.2 Systems of First-Order ODEs

Any higher-order ODE can be written as a system of first-order equations. This is the **state-space** representation:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (1.38)$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the input vector, and $\mathbf{A}$, $\mathbf{B}$ are system matrices.

The solution is:

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B}\mathbf{u}(\tau) d\tau \quad (1.39)$$

where $e^{\mathbf{A}t}$ is the **matrix exponential** (discussed in Section 1.6).

1.4 Laplace Transforms

The Laplace transform converts differential equations into algebraic equations, making system analysis much simpler.

1.4.1 Definition

Definition 0.2: Laplace Transform

The **Laplace transform** of a function $f(t)$ defined for $t \geq 0$ is:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st} dt \quad (1.40)$$

where $s = \sigma + j\omega$ is a complex variable.

The **inverse Laplace transform** is:

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds \quad (1.41)$$

In practice, we use tables and partial fraction expansion rather than direct integration.

Signal Name	$f(t)$ for $t \geq 0$	$F(s)$
Impulse	$\delta(t)$	1
Unit step	$u(t) = 1$	$\frac{1}{s}$
Ramp	t	$\frac{1}{s^2}$
Parabolic	t^2	$\frac{2}{s^3}$
Power	t^n	$\frac{n!}{s^{n+1}}$
Exponential	e^{-at}	$\frac{1}{s+a}$
Decaying ramp	te^{-at}	$\frac{1}{(s+a)^2}$
Sine	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
Cosine	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
Damped sine	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
Damped cosine	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Table 1.3: Common Laplace transform pairs.

Property	Time Domain	s -Domain
Linearity	$af(t) + bg(t)$	$aF(s) + bG(s)$
Differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
Second derivative	$\frac{d^2f}{dt^2}$	$s^2F(s) - sf(0^-) - f'(0^-)$
n th derivative	$\frac{d^nf}{dt^n}$	$s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^-)$
Integration	$\int_0^t f(\tau) d\tau$	$\frac{F(s)}{s}$
Time delay	$f(t - \tau)u(t - \tau)$	$e^{-\tau s} F(s)$
Frequency shift	$e^{-at} f(t)$	$F(s + a)$
Time scaling	$f(at)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$
Convolution	$f(t) * g(t)$	$F(s)G(s)$
Initial value	$\lim_{t \rightarrow 0^+} f(t)$	$\lim_{s \rightarrow \infty} sF(s)$
Final value	$\lim_{t \rightarrow \infty} f(t)$	$\lim_{s \rightarrow 0} sF(s)$

Table 1.4: Laplace transform properties.

1.4.2 Common Laplace Transform Pairs

1.4.3 Laplace Transform Properties

Important Note

The **Final Value Theorem** only applies if all poles of $sF(s)$ are in the left half-plane (LHP). If any poles are on the imaginary axis or in the right half-plane, the theorem gives incorrect results.

1.4.4 Partial Fraction Expansion

To find the inverse Laplace transform, we decompose rational functions into simpler terms.

Distinct Real Poles:

$$\frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)} = \frac{A_1}{s+p_1} + \frac{A_2}{s+p_2} + \cdots + \frac{A_n}{s+p_n} \quad (1.42)$$

The residue A_i is found by:

$$A_i = \lim_{s \rightarrow -p_i} (s+p_i)F(s) \quad (1.43)$$

Repeated Poles: For a pole of multiplicity m at $s = -p$:

$$\frac{N(s)}{(s+p)^m} = \frac{A_1}{s+p} + \frac{A_2}{(s+p)^2} + \cdots + \frac{A_m}{(s+p)^m} \quad (1.44)$$

Complex Conjugate Poles: For poles at $s = -\sigma \pm j\omega$:

$$\frac{N(s)}{(s+\sigma)^2 + \omega^2} = \frac{As+B}{(s+\sigma)^2 + \omega^2} \quad (1.45)$$

Example 0.3: Partial Fraction Expansion

Find the inverse Laplace transform of:

$$F(s) = \frac{2s+3}{(s+1)(s+2)} \quad (1.46)$$

Solution:

$$F(s) = \frac{A}{s+1} + \frac{B}{s+2} \quad (1.47)$$

Residues:

$$A = \lim_{s \rightarrow -1} (s+1) \cdot \frac{2s+3}{(s+1)(s+2)} = \frac{2(-1)+3}{-1+2} = 1 \quad (1.48)$$

$$B = \lim_{s \rightarrow -2} (s+2) \cdot \frac{2s+3}{(s+1)(s+2)} = \frac{2(-2)+3}{-2+1} = 1 \quad (1.49)$$

Therefore:

$$f(t) = e^{-t} + e^{-2t}, \quad t \geq 0 \quad (1.50)$$

1.4.5 Transfer Functions

Definition 0.3: Transfer Function

The **transfer function** $G(s)$ of a linear time-invariant system is the ratio of the Laplace transform of the output to the Laplace transform of the input, assuming zero initial conditions:

$$G(s) = \frac{Y(s)}{U(s)} \quad (1.51)$$

Transfer functions are typically written as:

$$G(s) = \frac{b_ms^m + b_{m-1}s^{m-1} + \cdots + b_1s + b_0}{a_ns^n + a_{n-1}s^{n-1} + \cdots + a_1s + a_0} = K \frac{\prod_{i=1}^m (s - z_i)}{\prod_{j=1}^n (s - p_j)} \quad (1.52)$$

where z_i are the **zeros** and p_j are the **poles** of the transfer function.

1.5 Fourier Transform and Frequency Analysis

1.5.1 Fourier Series

Any periodic function $f(t)$ with period T can be expressed as:

$$f(t) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2\pi nt}{T}\right) + b_n \sin\left(\frac{2\pi nt}{T}\right) \right] \quad (1.53)$$

where:

$$a_0 = \frac{1}{T} \int_0^T f(t) dt \quad (1.54)$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos\left(\frac{2\pi nt}{T}\right) dt \quad (1.55)$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin\left(\frac{2\pi nt}{T}\right) dt \quad (1.56)$$

Complex exponential form:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T} \quad (1.57)$$

$$\text{where } c_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

1.5.2 Fourier Transform

For non-periodic functions, the Fourier transform extends the series to continuous frequencies:

Definition 0.4: Fourier Transform

$$F(\omega) = \mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt \quad (1.58)$$

$$f(t) = \mathcal{F}^{-1}\{F(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{j\omega t} d\omega \quad (1.59)$$

Relationship to Laplace Transform: For causal signals, $F(\omega) = F(s)|_{s=j\omega}$ when the Laplace transform converges on the imaginary axis.

1.5.3 Frequency Response

For a stable LTI system with transfer function $G(s)$, the **frequency response** is:

$$G(j\omega) = |G(j\omega)|e^{j\angle G(j\omega)} \quad (1.60)$$

- $|G(j\omega)|$ = magnitude (gain) at frequency ω
- $\angle G(j\omega)$ = phase shift at frequency ω

For a sinusoidal input $u(t) = A \sin(\omega t)$, the steady-state output is:

$$y_{ss}(t) = A|G(j\omega)| \sin(\omega t + \angle G(j\omega)) \quad (1.61)$$

1.5.4 Bode Plots

Bode plots display magnitude (in dB) and phase versus frequency (on a log scale):

$$|G(j\omega)|_{\text{dB}} = 20 \log_{10} |G(j\omega)| \quad (1.62)$$

1.6 Linear Algebra

Linear algebra is essential for state-space control, multi-input multi-output (MIMO) systems, and machine learning.

1.6.1 Vectors and Matrices

Vector Operations:

- Addition: $\mathbf{x} + \mathbf{y} = [x_1 + y_1, x_2 + y_2, \dots, x_n + y_n]^\top$
- Scalar multiplication: $\alpha \mathbf{x} = [\alpha x_1, \alpha x_2, \dots, \alpha x_n]^\top$
- Dot product: $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i$
- Norm: $\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}$ (Euclidean norm)

Matrix Operations:

- Addition: $(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$

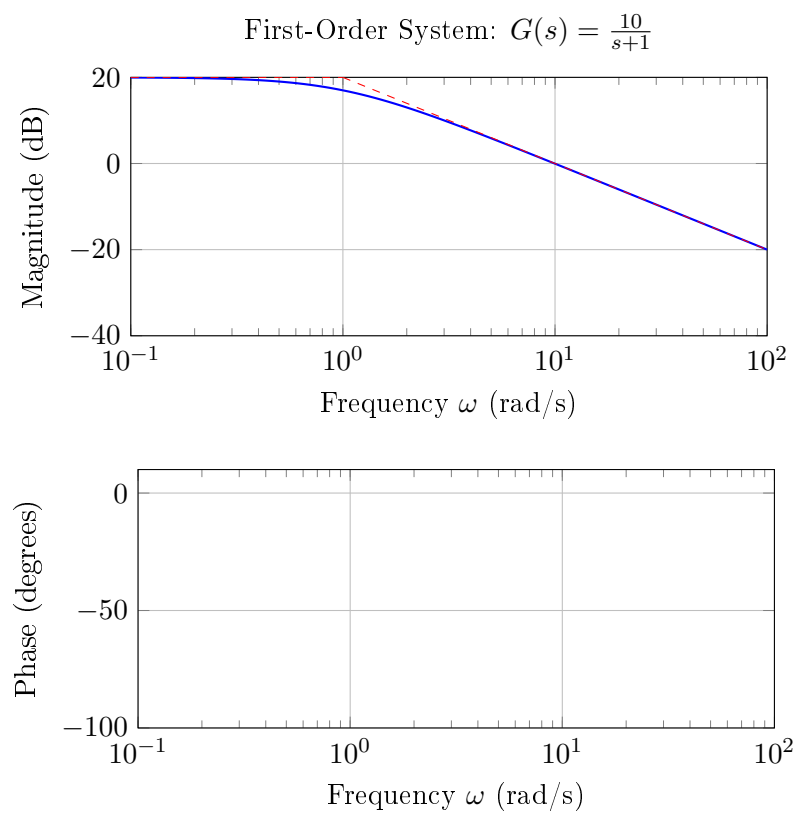


Figure 1.3: Bode plot for a first-order system.

- Multiplication: $(\mathbf{AB})_{ij} = \sum_k a_{ik}b_{kj}$
- Transpose: $(\mathbf{A}^\top)_{ij} = a_{ji}$
- Inverse: $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ (if exists)

1.6.2 Special Matrices

- **Identity matrix:** $\mathbf{I}$ with ones on diagonal, zeros elsewhere
- **Diagonal matrix:** $\mathbf{D} = \text{diag}(d_1, d_2, \dots, d_n)$
- **Symmetric matrix:** $\mathbf{A} = \mathbf{A}^\top$
- **Positive definite:** $\mathbf{x}^\top \mathbf{Ax} > 0$ for all $\mathbf{x} \neq \mathbf{0}$
- **Orthogonal matrix:** $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$

1.6.3 Determinant and Rank

The **determinant** of a square matrix:

$$\det(\mathbf{A}) = |\mathbf{A}| = \sum_{j=1}^n (-1)^{1+j} a_{1j} M_{1j} \quad (1.63)$$

where M_{1j} is the minor (determinant of submatrix).

Properties:

- $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$
- $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$
- $\det(\mathbf{A}^\top) = \det(\mathbf{A})$
- $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$

The **rank** is the number of linearly independent rows (or columns):

$$\text{rank}(\mathbf{A}) \leq \min(m, n) \text{ for } \mathbf{A} \in \mathbb{R}^{m \times n} \quad (1.64)$$

1.6.4 Eigenvalues and Eigenvectors

Definition 0.5: Eigenvalue Problem

For a square matrix $\mathbf{A}$, scalar λ is an **eigenvalue** and non-zero vector $\mathbf{v}$ is the corresponding **eigenvector** if:

$$\mathbf{Av} = \lambda \mathbf{v} \quad (1.65)$$

Eigenvalues are roots of the **characteristic equation**:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0 \quad (1.66)$$

Properties:

- Sum of eigenvalues = $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$
- Product of eigenvalues = $\det(\mathbf{A})$
- Eigenvalues of $\mathbf{A}^{-1}$ are $1/\lambda_i$
- Eigenvalues of $\mathbf{A}^n$ are λ_i^n

Stability and Eigenvalues

A continuous-time system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ is:

- **Stable** if all eigenvalues of $\mathbf{A}$ have negative real parts
- **Marginally stable** if eigenvalues have non-positive real parts with simple imaginary-axis eigenvalues
- **Unstable** if any eigenvalue has positive real part

1.6.5 Matrix Decompositions

Eigendecomposition: For diagonalizable matrix $\mathbf{A}$:

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} \quad (1.67)$$

where $\mathbf{V}$ contains eigenvectors and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Singular Value Decomposition (SVD): Any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \quad (1.68)$$

where $\mathbf{U}, \mathbf{V}$ are orthogonal and $\mathbf{\Sigma}$ contains singular values.

QR Decomposition:

$$\mathbf{A} = \mathbf{Q}\mathbf{R} \quad (1.69)$$

where $\mathbf{Q}$ is orthogonal and $\mathbf{R}$ is upper triangular.

LU Decomposition:

$$\mathbf{A} = \mathbf{L}\mathbf{U} \quad (1.70)$$

where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.

1.6.6 Matrix Exponential

The matrix exponential is crucial for solving state-space equations:

Definition 0.6: Matrix Exponential

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2!} + \frac{(\mathbf{A}t)^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{(\mathbf{A}t)^k}{k!} \quad (1.71)$$

Properties:

- $e^0 = \mathbf{I}$

- $\frac{d}{dt}e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t}$
- $e^{\mathbf{A}(t_1+t_2)} = e^{\mathbf{A}t_1}e^{\mathbf{A}t_2}$
- $(e^{\mathbf{A}t})^{-1} = e^{-\mathbf{A}t}$

Computation via Laplace:

$$e^{\mathbf{A}t} = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\} \quad (1.72)$$

Computation via eigendecomposition: If $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$:

$$e^{\mathbf{A}t} = \mathbf{V}e^{\mathbf{\Lambda}t}\mathbf{V}^{-1} = \mathbf{V} \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})\mathbf{V}^{-1} \quad (1.73)$$

1.6.7 Linear Systems of Equations

Solving $\mathbf{A}\mathbf{x} = \mathbf{b}$:

- **Unique solution** if $\mathbf{A}$ is square and invertible: $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$
- **No solution** if $\mathbf{b} \notin \text{range}(\mathbf{A})$
- **Infinite solutions** if $\text{null}(\mathbf{A}) \neq \{\mathbf{0}\}$

Least Squares: When $\mathbf{A}\mathbf{x} = \mathbf{b}$ has no exact solution:

$$\hat{\mathbf{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} = \mathbf{A}^\dagger \mathbf{b} \quad (1.74)$$

where $\mathbf{A}^\dagger$ is the Moore-Penrose pseudoinverse.

1.7 Probability and Statistics

Probability theory is essential for stochastic control, Kalman filtering, and machine learning.

1.7.1 Probability Fundamentals

Axioms of Probability:

1. $P(A) \geq 0$ for all events A
2. $P(\Omega) = 1$ where Ω is the sample space
3. For mutually exclusive events: $P(A \cup B) = P(A) + P(B)$

Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (1.75)$$

Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (1.76)$$

1.7.2 Random Variables

Discrete Random Variables:

- Probability mass function: $p(x) = P(X = x)$
- Expected value: $E[X] = \sum_x x \cdot p(x)$
- Variance: $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$

Continuous Random Variables:

- Probability density function: $f(x)$ where $P(a \leq X \leq b) = \int_a^b f(x) dx$
- Expected value: $E[X] = \int_{-\infty}^{\infty} x f(x) dx$
- Cumulative distribution: $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$

1.7.3 Common Distributions

Gaussian (Normal) Distribution:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (1.77)$$

Notation: $X \sim \mathcal{N}(\mu, \sigma^2)$

Multivariate Gaussian:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) \quad (1.78)$$

where $\boldsymbol{\mu}$ is the mean vector and Σ is the covariance matrix.

Other Important Distributions:

- **Uniform:** $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$
- **Exponential:** $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$
- **Bernoulli:** $P(X = 1) = p$, $P(X = 0) = 1 - p$
- **Poisson:** $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

1.7.4 Stochastic Processes

A **stochastic process** is a collection of random variables indexed by time: $\{X(t) : t \in T\}$

White Noise: Uncorrelated random sequence with:

- $E[w(t)] = 0$
- $E[w(t)w(\tau)] = \sigma^2 \delta(t - \tau)$

Markov Process: Future states depend only on present state:

$$P(X_{n+1} | X_n, X_{n-1}, \dots, X_0) = P(X_{n+1} | X_n) \quad (1.79)$$

1.7.5 Estimation Theory

Maximum Likelihood Estimation (MLE):

$$\hat{\theta}_{ML} = \arg \max_{\theta} \prod_{i=1}^n f(x_i|\theta) = \arg \max_{\theta} \sum_{i=1}^n \log f(x_i|\theta) \quad (1.80)$$

Bayesian Estimation:

$$p(\theta|\mathbf{x}) \propto p(\mathbf{x}|\theta)p(\theta) \quad (1.81)$$

(posterior $\propto$ likelihood $\times$ prior)

Minimum Mean Square Error (MMSE):

$$\hat{\theta}_{MMSE} = E[\theta|\mathbf{x}] \quad (1.82)$$

1.8 Optimization

Optimization is central to optimal control, parameter estimation, and machine learning.

1.8.1 Unconstrained Optimization

Necessary Condition: At a local minimum $\mathbf{x}^*$:

$$\nabla f(\mathbf{x}^*) = \mathbf{0} \quad (1.83)$$

Sufficient Condition: Also $\mathbf{H}(\mathbf{x}^*) \succ 0$ (positive definite Hessian).

Gradient Descent:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k) \quad (1.84)$$

where $\alpha > 0$ is the step size (learning rate).

Newton's Method:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - [\mathbf{H}(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k) \quad (1.85)$$

1.8.2 Constrained Optimization

Equality Constraints: Minimize $f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) = \mathbf{0}$

The **Lagrangian**:

$$L(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \boldsymbol{\lambda}^\top \mathbf{g}(\mathbf{x}) \quad (1.86)$$

Necessary Conditions (KKT):

$$\nabla_{\mathbf{x}} L = \mathbf{0} \quad (1.87)$$

$$\nabla_{\boldsymbol{\lambda}} L = \mathbf{g}(\mathbf{x}) = \mathbf{0} \quad (1.88)$$

Inequality Constraints: Minimize $f(\mathbf{x})$ subject to $\mathbf{h}(\mathbf{x}) \leq \mathbf{0}$

The **Karush-Kuhn-Tucker (KKT) conditions**:

$$\nabla f(\mathbf{x}^*) + \sum_i \mu_i \nabla h_i(\mathbf{x}^*) = \mathbf{0} \quad (1.89)$$

$$h_i(\mathbf{x}^*) \leq 0 \quad (1.90)$$

$$\mu_i \geq 0 \quad (1.91)$$

$$\mu_i h_i(\mathbf{x}^*) = 0 \quad (\text{complementary slackness}) \quad (1.92)$$

1.8.3 Convex Optimization

Definition 0.7: Convex Function

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if for all $\mathbf{x}, \mathbf{y}$ and $\theta \in [0, 1]$:

$$f(\theta\mathbf{x} + (1 - \theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1 - \theta)f(\mathbf{y}) \quad (1.93)$$

Key property: Any local minimum of a convex function is a global minimum.

Examples of Convex Problems:

- Linear programming: $\min \mathbf{c}^\top \mathbf{x}$ s.t. $\mathbf{Ax} \leq \mathbf{b}$
- Quadratic programming: $\min \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x}$ s.t. $\mathbf{Ax} \leq \mathbf{b}$ (if $\mathbf{Q} \succeq 0$)
- Least squares: $\min \|\mathbf{Ax} - \mathbf{b}\|^2$

1.8.4 Dynamic Programming

For sequential decision problems, **Bellman's Principle of Optimality**:

An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

The **Bellman equation** for discrete-time optimal control:

$$V^*(x_k) = \min_{u_k} [L(x_k, u_k) + V^*(f(x_k, u_k))] \quad (1.94)$$

where V^* is the optimal value function (cost-to-go).

1.9 Discrete-Time Mathematics

Digital control systems operate on sampled data, requiring discrete-time analysis.

1.9.1 Sequences and Difference Equations

A **sequence** is a function defined on integers: $x[n]$ or x_n for $n \in \mathbb{Z}$.

First-order difference equation:

$$x[n+1] = ax[n] + bu[n] \quad (1.95)$$

Solution:

$$x[n] = a^n x[0] + \sum_{k=0}^{n-1} a^{n-1-k} bu[k] \quad (1.96)$$

1.9.2 The Z-Transform

Definition 0.8: Z-Transform

The **Z-transform** of a sequence $x[n]$ is:

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=0}^{\infty} x[n]z^{-n} \quad (1.97)$$

Common Z-Transform Pairs:

Sequence	$x[n]$ for $n \geq 0$	$X(z)$
Unit impulse	$\delta[n]$	1
Unit step	$u[n] = 1$	$\frac{z}{z-1}$
Ramp	n	$\frac{z}{(z-1)^2}$
Exponential	a^n	$\frac{z}{z-a}$
Exponential ramp	na^n	$\frac{az}{(z-a)^2}$
Sine	$\sin(\omega n)$	$\frac{z \sin \omega}{z^2 - 2z \cos \omega + 1}$
Cosine	$\cos(\omega n)$	$\frac{z(z - \cos \omega)}{z^2 - 2z \cos \omega + 1}$

Table 1.5: Common Z-transform pairs.

Z-Transform Properties:

Property	Sequence	Z-Domain
Linearity	$ax[n] + by[n]$	$aX(z) + bY(z)$
Time shift (delay)	$x[n - k]$	$z^{-k}X(z)$
Time advance	$x[n + 1]$	$zX(z) - zx[0]$
Convolution	$x[n] * y[n]$	$X(z)Y(z)$
Initial value	$x[0]$	$\lim_{z \rightarrow \infty} X(z)$
Final value	$\lim_{n \rightarrow \infty} x[n]$	$\lim_{z \rightarrow 1} (z - 1)X(z)$

Table 1.6: Z-transform properties.

1.9.3 Relationship Between s-Domain and z-Domain

For a sampled signal with sampling period T :

$$z = e^{sT} \quad (1.98)$$

Therefore:

$$s = \frac{1}{T} \ln z \quad (1.99)$$

Stability in z-Domain

A discrete-time system is stable if and only if all poles of its transfer function $G(z)$ lie **inside the unit circle** $|z| < 1$.

The imaginary axis in the s -plane maps to the unit circle in the z -plane:

- Left half-plane ($\text{Re}(s) < 0$) $\rightarrow$ inside unit circle ($|z| < 1$)
- Right half-plane ($\text{Re}(s) > 0$) $\rightarrow$ outside unit circle ($|z| > 1$)
- $s = 0 \rightarrow z = 1$
- $s = \pm j\omega_s/2 \rightarrow z = -1$ (Nyquist frequency)

1.9.4 Discrete-Time State-Space

The discrete-time state-space representation:

$$\mathbf{x}[k+1] = \mathbf{A}_d \mathbf{x}[k] + \mathbf{B}_d \mathbf{u}[k] \quad (1.100)$$

$$\mathbf{y}[k] = \mathbf{C}_d \mathbf{x}[k] + \mathbf{D}_d \mathbf{u}[k] \quad (1.101)$$

Conversion from continuous-time (exact discretization with zero-order hold):

$$\mathbf{A}_d = e^{\mathbf{A}T} \quad (1.102)$$

$$\mathbf{B}_d = \int_0^T e^{\mathbf{A}\tau} d\tau \cdot \mathbf{B} = \mathbf{A}^{-1}(e^{\mathbf{A}T} - \mathbf{I})\mathbf{B} \quad (1.103)$$

1.10 Special Functions and Signals

1.10.1 Unit Step Function

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \quad (1.104)$$

The discrete-time unit step:

$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases} \quad (1.105)$$

1.10.2 Impulse (Dirac Delta) Function

The **Dirac delta function** is defined by:

$$\delta(t) = 0 \quad \text{for } t \neq 0 \quad (1.106)$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad (1.107)$$

Sifting property:

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0) \quad (1.108)$$

Relationship to step function:

$$\delta(t) = \frac{du(t)}{dt} \quad (1.109)$$

1.10.3 Ramp Function

$$r(t) = t \cdot u(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases} \quad (1.110)$$

1.10.4 Sinusoidal Functions

General sinusoid:

$$x(t) = A \sin(\omega t + \phi) = A \cos(\omega t + \phi - \pi/2) \quad (1.111)$$

where:

- A = amplitude
- $\omega = 2\pi f$ = angular frequency (rad/s)
- f = frequency (Hz)
- ϕ = phase (rad)
- $T = 1/f = 2\pi/\omega$ = period (s)

Complex exponential representation:

$$A \cos(\omega t + \phi) = \text{Re}\{Ae^{j(\omega t + \phi)}\} \quad (1.112)$$

1.10.5 Exponential Functions

Real exponential:

$$x(t) = Ae^{\sigma t} \quad (1.113)$$

- $\sigma < 0$: decaying
- $\sigma > 0$: growing
- $\sigma = 0$: constant

Complex exponential (damped sinusoid):

$$x(t) = Ae^{\sigma t} \cos(\omega t + \phi) \quad (1.114)$$

1.11 Numerical Methods

1.11.1 Numerical Integration of ODEs

Euler's Method (Forward):

$$x_{n+1} = x_n + hf(t_n, x_n) \quad (1.115)$$

Runge-Kutta 4th Order (RK4):

$$k_1 = f(t_n, x_n) \quad (1.116)$$

$$k_2 = f(t_n + h/2, x_n + hk_1/2) \quad (1.117)$$

$$k_3 = f(t_n + h/2, x_n + hk_2/2) \quad (1.118)$$

$$k_4 = f(t_n + h, x_n + hk_3) \quad (1.119)$$

$$x_{n+1} = x_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (1.120)$$

1.11.2 Root Finding

Newton-Raphson:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1.121)$$

Bisection Method: For continuous f with $f(a)f(b) < 0$, repeatedly halve the interval containing the root.

1.11.3 Numerical Linear Algebra

Key algorithms:

- **Gaussian elimination:** $O(n^3)$ for solving $\mathbf{Ax} = \mathbf{b}$
- **QR iteration:** Computing eigenvalues
- **SVD algorithms:** Golub-Kahan bidiagonalization
- **Iterative methods:** Conjugate gradient, GMRES for large sparse systems

1.12 Chapter Summary

Key Mathematical Concepts for Control Systems

1. **Complex Numbers:** Foundation for frequency analysis and pole-zero locations
2. **Calculus:** Describes system dynamics through derivatives and integrals
3. **Differential Equations:** Mathematical models of physical systems
4. **Laplace Transform:** Converts differential equations to algebraic equations
5. **Fourier Analysis:** Frequency-domain representation and analysis
6. **Linear Algebra:** Essential for state-space methods and MIMO systems
7. **Probability/Statistics:** Stochastic systems, estimation, and machine learning
8. **Optimization:** Controller design and parameter tuning
9. **Discrete Mathematics:** Digital control and sampled systems

1.13 Exercises

Exercise 1.1. Convert the following complex numbers to polar form:

- (a) $z_1 = -3 + 4j$
- (b) $z_2 = 2 - 2j$
- (c) $z_3 = -5$

Exercise 1.2. Find the Laplace transform of:

- (a) $f(t) = 3e^{-2t} \sin(4t)$
- (b) $f(t) = t^2 e^{-t}$
- (c) $f(t) = (1 - e^{-t})u(t)$

Exercise 1.3. Find the inverse Laplace transform of:

- (a) $F(s) = \frac{5}{s(s+3)}$
- (b) $F(s) = \frac{s+1}{s^2+4s+13}$
- (c) $F(s) = \frac{1}{(s+1)^2(s+2)}$

Exercise 1.4. For the matrix $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$:

- (a) Find the eigenvalues
- (b) Find the eigenvectors
- (c) Compute $e^{\mathbf{A}t}$
- (d) Is the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ stable?

Exercise 1.5. Linearize the nonlinear system $\dot{x} = x^2 - u$ about the equilibrium point where $x_0 = 1$, $u_0 = 1$.

Exercise 1.6. Find the Z-transform and region of convergence for:

- (a) $x[n] = (0.5)^n u[n]$
- (b) $x[n] = n(0.8)^n u[n]$
- (c) $x[n] = \cos(0.2\pi n)u[n]$

Exercise 1.7. Using gradient descent, find the minimum of $f(x, y) = x^2 + 2y^2 - 2xy - 2x$ starting from $(0, 0)$ with step size $\alpha = 0.1$. Perform 5 iterations.

Exercise 1.8. For a random variable $X \sim \mathcal{N}(5, 4)$:

- (a) Find $P(X > 7)$
- (b) Find $P(3 < X < 7)$
- (c) Find the value x such that $P(X < x) = 0.95$